

The effect of high animal protein intake on the risk of calcium stone-formation in the urinary tract.



1. Studies were carried out on six normal male subjects to determine the short-term effect of increasing the dietary consumption of animal protein on the urinary risk factors for stone-formation, namely, volume, pH, calcium oxalate, uric acid and glycosaminoglycans. 2. An increase of 34 g/day of animal protein in the diet significantly increased urinary calcium (23%) and oxalate (24%). Total urinary nitrogen increased by an average of 368 mmol/day. The accompanying increase in dietary purine (11 mmol of purine nitrogen/day) caused a 48% increase in the excretion of uric acid. 3. The overall relative probability of forming stones, calculated from a combination of the risk factors, was markedly increased (250%) throughout the period of high animal protein ingestion.